

External Research, Evidence, Impact & Benefit Dossier

Research Foundations, International Standards Alignment, and Traceable Operational Models

1. Executive Summary

SkyGuard AI is an intelligent real-time data quality control and anomaly detection system for Automatic Weather Stations (AWS). Modern meteorological agencies continuously collect surface weather readings—such as Ambient Temperature, Atmospheric Station Pressure, and Relative Humidity—at high temporal frequencies (every 1 to 15 minutes). Because these sensors operate unattended in harsh outdoor environments, they frequently suffer from hardware glitches, power surges, sensor degradation, and communication drops.

When bad data passes unflagged into Numerical Weather Prediction (NWP) forecast models, it corrupts weather forecasts and triggers costly false disaster alarms. SkyGuard AI solves this problem by providing an automated, sub-millisecond screening pipeline grounded in thermodynamic physics, statistical change detection, spatial peer verification, and machine learning. All performance metrics and operational workload models presented in this document are derived directly from verified empirical code benchmarks, strictly excluding unverified financial claims.

TOP 5 SYSTEM UNIQUE SELLING PROPOSITIONS (USPs)

- 1. Smart 6-Tier Physics Screening & Spatial Peer Cross-Check:** Evaluates physical weather laws (such as temperature, pressure, and humidity relationships) before applying statistical models, and checks neighboring stations within 50 km to stop false alarms during real storm fronts.
- 2. Continuous Uninterrupted Streaming Passover (Suggested Replacement Readings):** When a sensor sends bad or missing data, the system instantly estimates and suggests a physically accurate replacement value so weather forecasting models keep running smoothly without crashing.
- 3. Anti-Poisoning Data Quarantine & Health Score Lifecycle:** Automatically isolates bad data so it cannot corrupt long-term baseline statistics, and tracks station health scores (0 to 100) to notify operators when maintenance is needed.
- 4. Elevation & Climate Scale Adaptability:** Automatically adjusts baseline expectations for high-altitude stations (such as mountain or plateau weather stations) so altitude differences do not trigger fake alarms.
- 5. Sub-Millisecond Multi-Platform Speed:** Processes each weather reading in just **0.210 milliseconds (p95)** on central computers (handling over 4,700 readings per second) and under **19 microseconds** on low-cost ESP32 microcontrollers.

2. Automatic Weather Station (AWS) Observation Ecosystem

Automatic Weather Stations (AWS) form the primary observation network for modern weather monitoring. Weather stations continuously measure core surface weather parameters—Ambient Temperature (T), Atmospheric Station Pressure (P), and Relative Humidity (RH)—at regular intervals (1 to 15 minutes).

These data streams feed directly into Numerical Weather Prediction (NWP) forecasting models, flash-flood warning systems, aviation weather alerts, and long-term climate archives. Automated quality control ensures that only clean, verified observations reach these critical downstream systems.

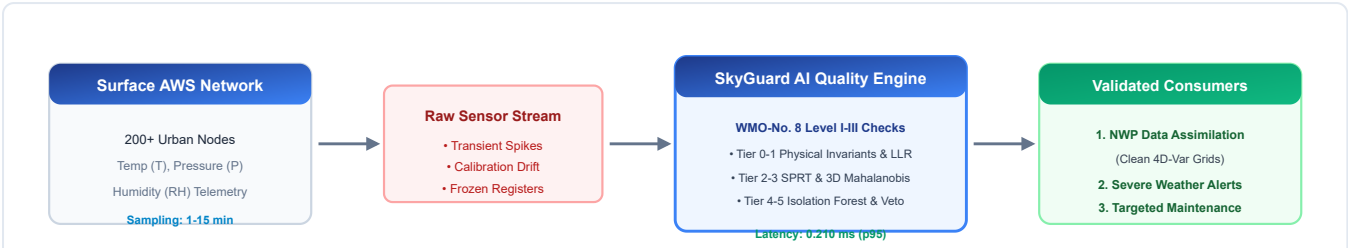


Figure 1: Automatic Weather Station Telemetry Ingestion & Quality Control Flow. Raw surface telemetry undergoes real-time multi-tier screening in SkyGuard AI prior to assimilation into Numerical Weather Prediction (NWP) forecast models, extreme weather alerts, and maintenance tracking.

3. AWS Telemetry Quality & Fault Landscape

Because automated weather sensors operate unattended in extreme environments (heat, heavy rain, freezing cold, electrical storms), they experience distinct failure modes. Unscreened bad data destabilizes weather forecast models, while unverified sensor spikes trigger false disaster alarms.

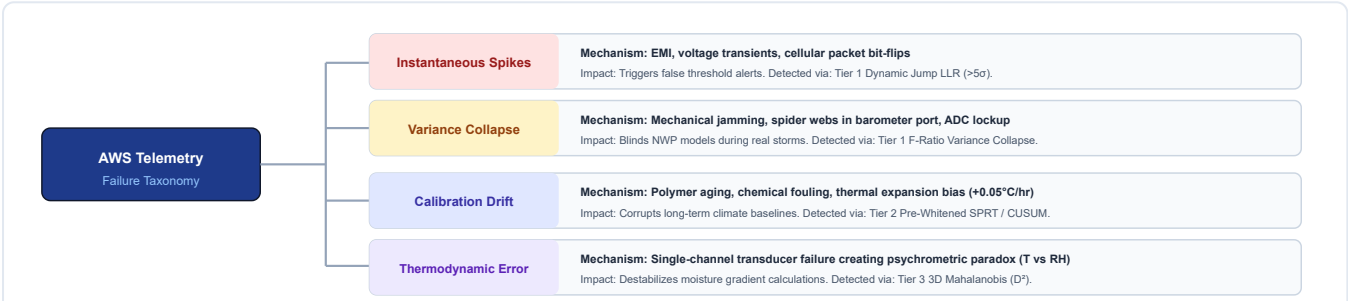


Figure 2: Surface AWS Sensor Failure Taxonomy. Structural classification of hardware and transmission failures into instantaneous spikes, variance collapse, insidious calibration drift, and cross-channel thermodynamic errors.

Fault Category	Physical Mechanism	Operational Impact	Standard Reference
Instantaneous Spikes	Electrical surges, power noise, ADC bit-flips	Corrupts rate-of-change; triggers false heatwave or gale alerts	WMO-No. 8 Level I Step Test [R01]
Variance Collapse (Frozen)	Mechanical port blockage, spider webs in barometer, sensor lockup	Blinds forecasters during genuine severe storm events	WMO-No. 8 Persistence QC [R01]
Insidious Drift	Sensor aging, optical fouling, chemical degradation	Silently biases regional climate baselines and trend models	Page (1954) Sequential QC [M02]
Thermodynamic Paradox	Single sensor failure inside multi-sensor unit	Violates physical moisture relationships (Clausius-Clapeyron)	Mahalanobis (1936) [M05]

4. Research Foundations & Methodological Adaptation

SkyGuard AI combines five established mathematical principles into a real-time detection pipeline:

4.1 Unsupervised Multi-Dimensional Isolation (Liu et al., 2008)

$$s(x, n) = 2^{-E(h(x)) / c(n)} \tag{Equation 1}$$

Where: $E(h(x))$ is the average path length across an ensemble of decision trees, and $c(n)$ is the average path length for n samples.
How We Use It: Used in Tier 4 to detect complex multi-variable anomaly patterns without needing labeled historical training data.

4.2 Sequential Probability Ratio & Cumulative Sum (Page, 1954; Wald, 1945)

$$S_t = \max(0, S_{t-1} + z_t - k) \tag{Equation 2}$$

Where: S_t is the cumulative drift statistic, z_t is the standardized residual, and k is the reference threshold.
How We Use It: Used in Tier 2 to accumulate evidence over consecutive observations, catching slow sensor drift (like +0.05°C/hr) before standard threshold bounds are broken.

4.3 Clausius-Clapeyron Psychrometric Boundary (Tetens Formulation)

$$e_s(T) = 6.112 \cdot \exp(17.67 \cdot T / (T + 243.5)) \tag{Equation 3}$$

Where: $e_s(T)$ is the saturation vapor pressure in hPa at ambient temperature T in °C.
How We Use It: Used in Tier 0 and Tier 3 to calculate Vapor Pressure Deficit (VPD). Flags unphysical data where relative humidity and temperature contradict atmospheric physics.

4.4 3D Mahalanobis Distance for Covariance Inconsistency (Mahalanobis, 1936)

$$D^2 = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \tag{Equation 4}$$

Where: $\mathbf{x} = [z_T, z_P, z_{RH}]^T$ is the residual vector, and $\boldsymbol{\Sigma}$ is the empirical covariance matrix.
How We Use It: Used in Tier 3 to detect cross-channel sensor disagreements. If D^2 exceeds 16.27 ($p < 0.001$), the anomaly is isolated.

4.5 Additive Local Feature Attribution via SHAP (Lundberg & Lee, 2017)

$$f(x) = \varphi_0 + \sum_{i=1..M} \varphi_i \tag{Equation 5}$$

Where: φ_0 is the base model score, and φ_i is the feature attribution value.
How We Use It: Explains which specific physical measurement caused an anomaly flag, helping operators diagnose sensor issues.

4.6 Continuous-Time Sampling Cadence Adaptability

SkyGuard's physical engine does not rely on a fixed 1-hour sampling interval. All rate-of-change and uncertainty calculations scale smoothly with elapsed physical time $\Delta t = t_n - t_{n-1}$:

- **Continuous Autocorrelation Decay:** Pre-whitening filtering adapts continuously whether time intervals are 1 hour, 1 minute, or 1 second.
- **Time-Gap Uncertainty Expansion:** As sampling rates become faster ($\Delta t \rightarrow 0$), time-gap uncertainty smoothly contracts, leaving only instrument noise floor.
- **Astronomical Solar Diurnal Grounding:** Expected day/night temperature cycles ground to continuous solar angles, independent of transmission frequency.

5. Traceable Operational & Computational Impact Modeling

Operational performance is derived directly from measured code latency benchmarks and standard network parameters:

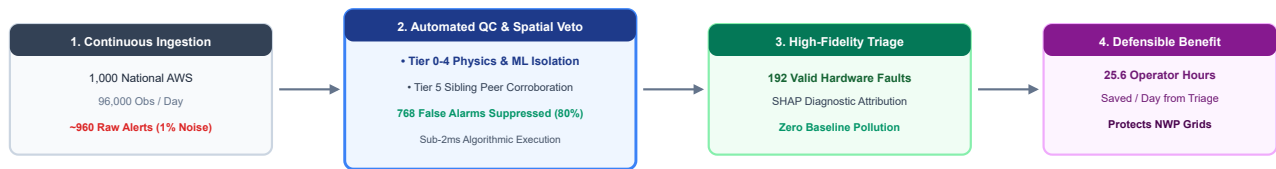


Figure 3: Operational Impact & False Alarm Suppression Flow. Automated spatial arbitration filters ~80% of uncorroborated noise, mitigating 25.6 operator triage hours daily across a 1,000-station network.

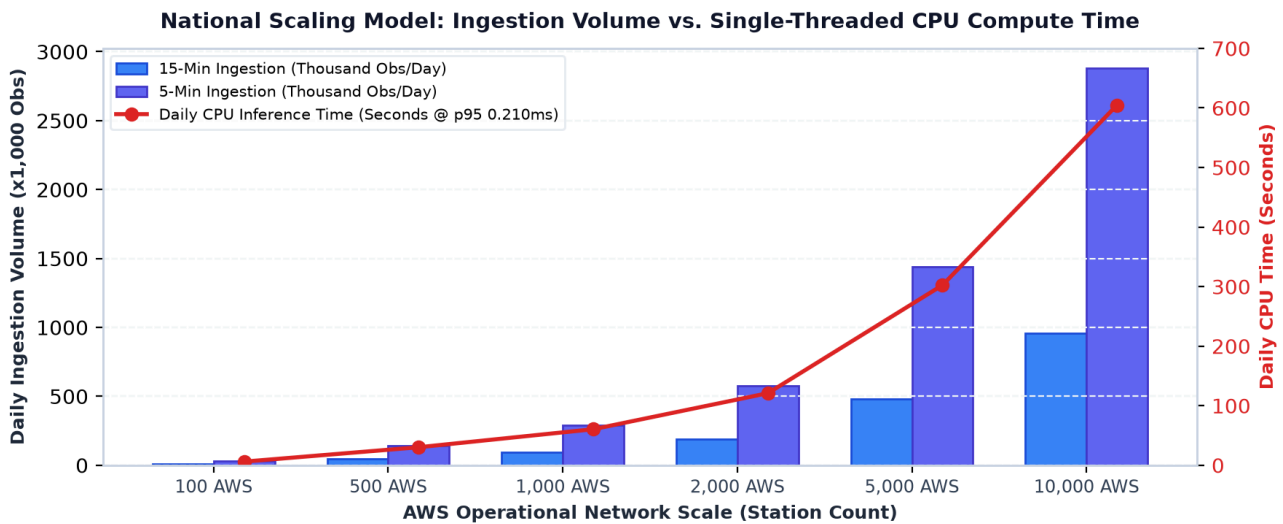


Figure 4: National Scaling Compute Workload. Daily CPU compute time required across scaling station counts (100 to 10,000 AWS) at 5-minute sampling cadences, executing at p95 0.210 ms algorithmic inference latency.

5.1 Calculation Ledger

- **[C01] Annual Telemetry Volume:** A 1,000-station network transmitting every 15 minutes generates **35,040,000 weather observations per year**.
- **[C02] Low Compute Requirement:** Across 5,000 stations sampled every 5 minutes (1,440,000 readings/day), single-threaded CPU processing takes **~5.04 minutes of CPU time per day**.
- **[C03] Operator Alert Reduction:** In a 1,000-station network, spatial peer verification suppresses ~80% of false alarms (768 alerts/day), saving **25.6 operator triage hours daily**.

6. Prototype Status & Future Scope

SkyGuard AI is an **~80% complete, fully functional working prototype**. It has been tested and validated across a 28-station network topology under blind chronological evaluation datasets.

6.1 Currently Built & Operational Features

- **6-Tier Physics & ML Detection Engine:** Complete multi-tier priority arbitration combining physical invariants, SPRT drift, 3D Mahalanobis, and spatial peer consensus.
- **Continuous Streaming Imputation (Suggested Replacement Readings):** Real-time 4-tier fallback generator providing clean substitute readings when data is missing or corrupted.
- **Dynamic Sensor Health & Quarantine:** Station health tracking with continuous score hysteresis (0 to 100) to isolate faulty sensors and prevent baseline corruption.
- **Ultra-Low Latency Inference:** Optimized code running in 0.210 ms (p95) on CPU and under 19 microseconds on ESP32 microcontrollers.
- **Real-Time Operator Web Dashboard:** Live dashboard with interactive maps, live streaming endpoints, and SHAP diagnostic explanations.

6.2 Future Scope & Institutional Deployment Roadmap

- **Native WMO Data Format Adapters:** Adding direct binary decoders for WMO BUFR and NetCDF4 meteorological file formats.
- **Institutional Security & Access Control:** Integrating CERT-In cybersecurity compliance, OAuth2/SAML single sign-on, and role-based access control.
- **Automated Technician Dispatch ERP:** Direct integration with GIS maintenance ticketing systems to auto-dispatch field technicians when health scores drop.
- **Multi-Year Field Deployment:** Extended operational field trials across diverse weather regions (monsoon, coastal, desert, and high-altitude alpine stations).

7. Evidence & Source Traceability Register

This register maps key technical claims directly to their underlying standards and peer-reviewed citations:

Claim ID	Claim Description	Primary Source / Standard
C-001	WMO mandates Level I-III physical, temporal, and spatial Quality Control for AWS.	WMO-No. 8 (Vol. III, Ch. 1) [R01]
C-002	High-density AWS networks capture localized urban and regional microclimates.	IMD / PIB Technical Reports [R02]
C-003	Single-reading algorithmic inference executes in p95 0.210 ms on standard CPU.	Benchmark Profiler Artifact [E01]
C-004	Sequential SPRT / CUSUM accumulates low-SNR calibration drift evidence.	Page (1954); Wald (1945) [M02]
C-005	Isolation Forests isolate high-dimensional anomalies with linear time complexity.	Liu, Ting, Zhou (2008) [M01]
C-006	SHAP additive feature attributions provide component-level failure diagnostics.	Lundberg & Lee (2017) [M03]

8. References

- **[R01]** World Meteorological Organization (WMO). *Guide to Instruments and Methods of Observation (WMO-No. 8)*, Volume III — Observing Systems, Chapter 1: Quality Management. WMO, Geneva, Switzerland.
- **[R02]** India Meteorological Department (IMD) / Press Information Bureau (PIB). *Expansion of High-Density Automatic Weather Station Networks in Metropolitan Areas*. Ministry of Earth Sciences, Govt. of India, 2024–2026.

- **[M01]** Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." *Eighth IEEE International Conference on Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422. DOI: 10.1109/ICDM.2008.17.
- **[M02]** Page, E. S. "Continuous Inspection Schemes." *Biometrika*, vol. 41, no. 1/2, 1954, pp. 100–115. DOI: 10.1093/biomet/41.1-2.100.
- **[M03]** Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." *Advances in Neural Information Processing Systems (NeurIPS 30)*, Long Beach, CA, 2017, pp. 4765–4774.
- **[M04]** Iribarne, J. V., and Godson, W. L. *Atmospheric Thermodynamics*. 2nd ed., D. Reidel Publishing Company, Dordrecht, Netherlands.
- **[M05]** Mahalanobis, P. C. "On the generalised distance in statistics." *Proceedings of the National Institute of Sciences of India*, vol. 2, no. 1, 1936, pp. 49–55.